Poker AI: methods, evidence and current limits

Research snapshot | November 3, 2026

A validated simulation and game-theory project. The current global Hold'em policy remains a weak research prototype; no competitive strength or profitability has been established.

Own implementation, separate checks

The production path uses own card masks, five-to-seven-card evaluation, a 2-6-seat immutable no-limit engine, card/action abstractions, CFR variants and sampled CFR. Explicit random generators and versioned configs define all stochastic experiments. Frozen average strategies are evaluated after training; reference outputs never enter own regret tables, policies or checkpoints.

Production path	Purpose
Own mechanics -> own learning	Cards, legal betting, side pots, own self-play updates
Frozen own policies -> evaluation	Exact restricted best responses and paired duplicate matches
OpenSpiel -> comparison reports only	Independent numerical solver/value checks; shared own finite-game export

Verified foundation and earlier measurements

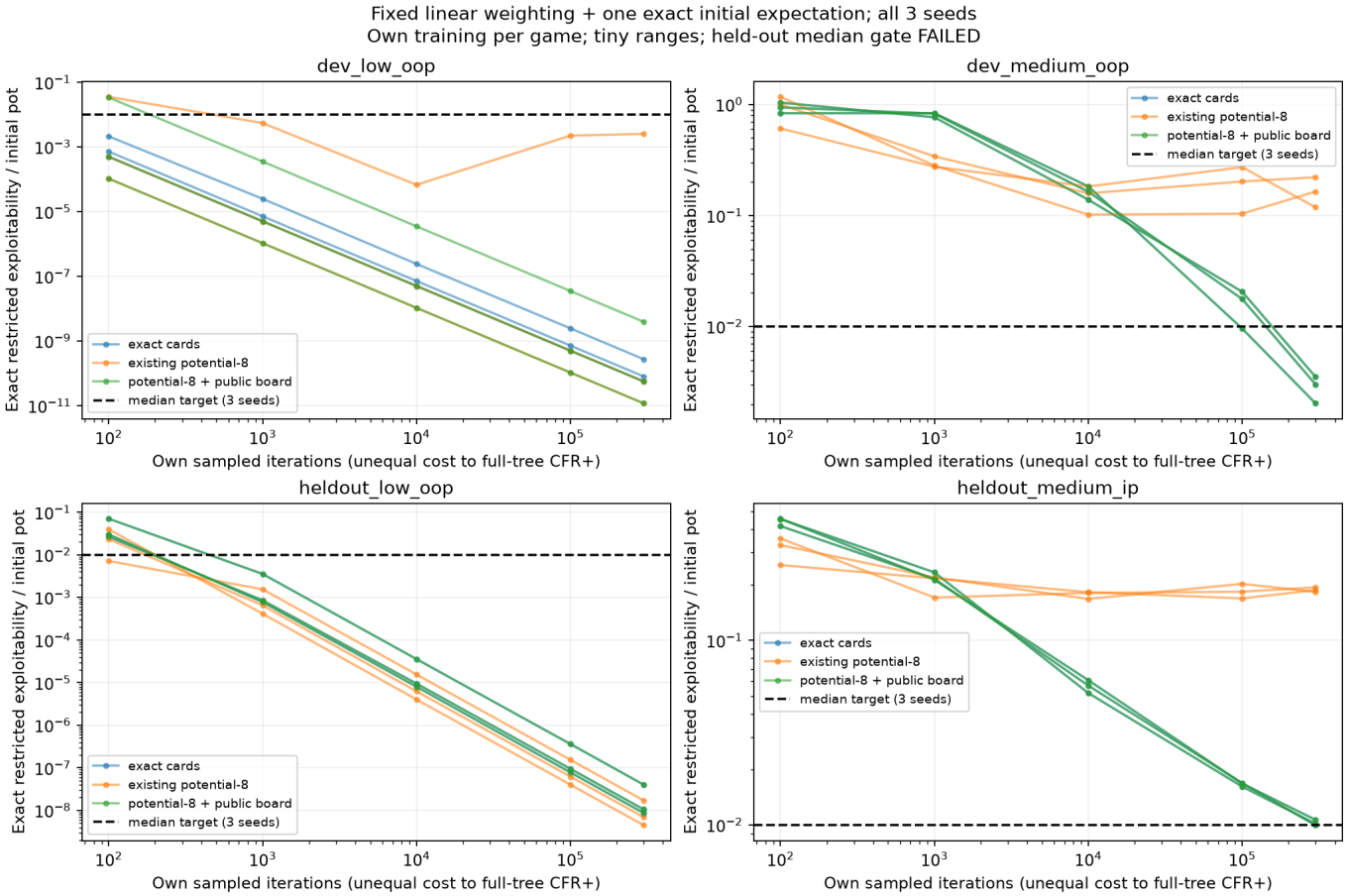
Area	Recorded evidence	What it establishes
Evaluator	2,598,960 exhaustive five-card hands; 2,000,000 sampled six/seven-card hands	Agreement with independent Treys ranks on those inputs
Exact toy solvers	Kuhn and Leduc; maximum matched exploitability difference 1.39e-16 ante chips/hand	Numerical agreement for CFR variants and specified games
Local global-policy evaluation	36,000 hands; 1,000 shared independent deck clusters; 0/6 positive primary tests	No confirmed positive advantage under the preset conservative family test
Selected extensions	360 exact Bayesian profiles; 5/5 empirical confidence-robust risk fractions are zero before fees	Finite toy-model security checks and conditional risk analysis; no real-world edge

Evaluation uses both seats and three frozen training seeds within each deck cluster. Thus 36,000 hands are not 36,000 independent observations. The primary Bonferroni-Hoeffding rule is conservative; failure to reject does not prove zero advantage. Loss point estimates against tight-aggressive and equity bots remain visible.

What the new Hold'em diagnostic found

Four turn-river games, two physical combos per seat, three fixed seeds. Each game receives its own self-play training; this does not test zero-shot board generalization.

The preselected procedure uses linear regret/average weighting plus one exact expectation of the initial uniform policy. All four initialization-by-weighting controls completed at the same fixed 300,000 iterations. Full-tree CFR+ is a separate own accuracy control; sampled and full-tree iterations have unequal costs.



Lower is better; logarithmic axes. Blue: exact cards. Orange: existing potential-8 keys. Green: buckets plus observed public board. Each color has three seed curves; identical curves overlap. The dashed line is the preset median target.

Game	Exact median / pot	Existing buckets	Buckets + public board
Development low OOP	7.9850837e-11	5.5817288e-11	5.5817288e-11
Development medium OOP	0.003008241	0.16430114	0.003008241
Held-out low OOP	1.0519168e-08	6.9347915e-09	1.0519168e-08
Held-out medium IP	0.010152859	0.18719347	0.010152859

Held-out gate FAILED: median 0.010152858879237701 pots exceeds the unchanged 0.01 limit. The worst seed, 0.010664524586617995, passes its separate 0.025 limit. This is a narrow miss, not a pass after rounding. No global follow-up experiment was launched.

Failure analysis and honest next steps

Representation is a measured bottleneck

Existing keys merge distinct observed public boards. Preserving the board reduces median restricted exploitability from 16.43% to 0.30% of the pot on development medium and from 18.72% to 1.0153% on held-out medium. In both medium games the refined encoding becomes as fine as exact cards, so matching trajectories is expected. This local control does not establish scalable global generalization. Low-SPR results are not uniformly ordered by encoding.

Coverage and arithmetic require separate audits

One exact initial expectation gives every average row a real update. It does not mean every row was sampled afterward: as many as 827 development-medium rows remain initial-only. Reports separate these rows, sampled visits and initialization work. This full-tree operation is feasible only in the small diagnostics.

Initial native/own training trajectories diverged at near-zero tied regrets because fused multiply-add rounds differently from separate operations. Eight scalar contraction controls and an explicit optional Python/Numba kernel isolate that effect. Fixed-policy value/best-response checks already agreed. The failed original results were preserved; tolerances were not loosened.

Across 378 frozen policy records, native/own value error is at most 1.14e-13 chips and exploitability error at most 8.18e-14 chips, versus the preset 1e-9 tolerance. Sixty matched exact full-tree checkpoints have zero reported exploitability and mean-row-L1 disagreement on the recorded pinned build. These repeated policy records are not independent statistical samples.

Scope of the numbers

Exploitability here is exact restricted-game NashConv/2 divided by the initial pot. It is not full-game exploitability or a win rate. Cross-play EV uses a frozen own full-tree control; a fixed opponent may not expose weaknesses that a best response finds. The reference shares our mechanics/export. Earlier engine and evaluator comparisons provide separate mechanics evidence. Three seeds do not establish a population confidence interval; cross-hardware bit identity is not promised.

Reproduce and continue

From the repository root	Behavior
make check	Lint, format, strict typing and fast tests
make report-holdem-learning	Replay saved own policies; audit and regenerate tables/plots
make validate-holdem-learning	Retrain all fixed controls and hold-outs; nonzero exit when the learning gate fails

Next: validate an estimator that integrates the small river chance distribution, compare it at fixed node budgets, and preregister fresh held-out games before collecting revised results. All four current games are now known development evidence. Keep the same accuracy limits and retain every seed, then reconsider the global training comparison.

Evidence map (repository paths)

docs/holdem-strength-v2.md; results/holdem-learning/extended/; configs/holdem-learning-evidence.json; results/holdem-learning/report/summary.json. Earlier foundations: docs/phase1-verification.md, docs/phase2-solvers.md, docs/phase4-evaluation.md, docs/phase5-extensions.md. Reference licensing: docs/solver_validation.md. MCCFR method: Lanctot et al. (2009), cs.cmu.edu/~kwaugh/publications/nips09b.pdf.